Agentic 2-D Unstructured Compressible Finite-Volume Solver

Reproducible benchmark submission

August 28, 2026

Abstract

Eight supplied NACA0012 and cylinder cases were exercised with a distributed cell-centred finite-volume solver. The report distinguishes target convergence from bounded steady plateaus and records the Re=200 physical wake result: statistically periodic. Every plotted quantity is regenerated from the submitted CSV or VTK files.

1 Introduction

This benchmark evaluates a compressible, calorically perfect-gas, two-dimensional unstructured finite-volume solver on three inviscid NACA0012 cases, three laminar NACA0012 cases, a steady cylinder at Reynolds 20, and a vortex-shedding cylinder at Reynolds 200. Result status is taken from the solver metadata and is not inferred from a plot.

2 Meshes, Boundary Tags, and Case Inputs

The supplied CGNS files are read without case-specific geometry code. NACA cases use the airfoil mesh and cylinder cases use the cylinder mesh; boundary families are mapped from the JSON tags to farfield, slip-wall, or no-slip adiabatic-wall fluxes. The global mesh counts and tags below are taken from each submitted metadata/input pair.

Case	mesh	mode	M_∞	Re	global cells	global faces	boundary tags
naca0012_m015_inviscid	NACA0012_H2.cgns	inviscid	0.15	–	20816	36498	bc-2: farfield; bc-4: slip_wall
naca0012_m080_inviscid	NACA0012_H2.cgns	inviscid	0.8	–	20816	36498	bc-2: farfield; bc-4: slip_wall
naca0012_m200_inviscid	NACA0012_H2.cgns	inviscid	2.0	–	20816	36498	bc-2: farfield; bc-4: slip_wall
naca0012_m015_laminar_re5000	NACA0012_H2.cgns	laminar	0.15	5000.0	20816	36498	bc-2: farfield; bc-4: no_slip_adiabatic_wall
naca0012_m080_laminar_re5000	NACA0012_H2.cgns	laminar	0.8	5000.0	20816	36498	bc-2: farfield; bc-4: no_slip_adiabatic_wall
naca0012_m200_laminar_re5000	NACA0012_H2.cgns	laminar	2.0	5000.0	20816	36498	bc-2: farfield; bc-4: no_slip_adiabatic_wall
cylinder_m010_laminar_re20	CylinderB1.cgns	laminar	0.1	20.0	10185	20180	WALL: no_slip_adiabatic_wall; FAR: farfield
cylinder_m010_laminar_re200	CylinderB1.cgns	laminar	0.1	200.0	10185	20180	WALL: no_slip_adiabatic_wall; FAR: farfield

3 Governing Equations and Nondimensionalization

The conservative state is $\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T$ and $p = (\gamma-1)\rho e$, $E = e+(u^2+v^2)/2$, $a = \sqrt{\gamma p/\rho}$. The inviscid fluxes are $\mathbf{F}^i = [\rho u, \rho u^2+p, \rho uv, u(\rho E+p)]^T$ and $\mathbf{G}^i = [\rho v, \rho uv, \rho v^2+p, v(\rho E+p)]^T$. Face residuals use $\mathbf{R}_i = \sum_f(\mathbf{F}^i - \mathbf{F}^v)_f \ell_f$. Laminar viscosity is $\mu = \rho_\infty U_\infty L/Re$; Newtonian stress uses $\tau_{xx} = 2\mu u_x - \frac{2}{3}\mu(u_x+v_y)$, $\tau_{yy} = 2\mu v_y - \frac{2}{3}\mu(u_x+v_y)$, $\tau_{xy} = \mu(u_y+v_x)$, and Fourier heat flux $\mathbf{q} = -k\nabla T$ with $k = \mu c_p/Pr$. Coefficients use $q_\infty = \frac{1}{2}\rho_\infty U_\infty^2$ and the input reference area and length.

4 Spatial Discretization

The method is a cell-centred unstructured finite-volume scheme. Each face contributes its selected oriented numerical flux to the two adjacent owned residuals; boundary faces use the JSON family tag and a consistent outward normal. Primitive gradients are reconstructed with a least-squares stencil and used for piecewise-linear face states in every production result.

4.1 Second-Order Reconstruction

Least-squares gradients and piecewise-linear reconstruction are used in the final residual, force, surface, and field outputs. The Re=200 production mode is documented separately as a lagged BDF2 operator when selected; this is not silently relabelled as an active-refresh solve. A positivity backtrack reduces the reconstruction coefficient toward the cell average only when a candidate density or pressure would fall below the configured floor. The Re=200 production residual uses the active solution-dependent reconstruction at the accepted state.

4.2 Limiter and Positivity Control

The active Barth–Jespersen component limiter bounds reconstructed primitive values against the local stencil extrema. Conservative updates are positivity checked and reverted/backtracked when necessary, with the final field checked for positive density and pressure in the machine-readable sanity file. The Mach-2 laminar continuation used the explicit pressure floor $p_{\min} = 0.1$ recorded in its launch environment and metadata; this is a solver-side admissibility guard for the smallest trailing-edge shock-layer cells, not a post-processing clip.

4.3 Inviscid and Viscous Fluxes

The solver implements Roe's approximate Riemann flux with a quadratic Harten-style spectral smoothing when selected, and Rusanov/local Lax–Friedrichs otherwise. The actual per-case flux choice is read from metadata.json: Roe approximate Riemann with quadratic Harten-style entropy smoothing: naca0012_m015_inviscid, naca0012_m080_inviscid, naca0012_m200_inviscid; Rusanov local Lax-Friedrichs: naca0012_m015_laminar_re5000, naca0012_m080_laminar_re5000, naca0012_m200_laminar_re5000, cylinder_m010_laminar_re20, cylinder_m010_laminar_re200. Thus no entropy-fix claim is attributed to a Rusanov result. The Re=200 low-Mach Rusanov metadata records a pressure-jump acoustic fraction of 0.1 and a normal-momentum velocity-jump acoustic fraction of 0.1. Internal-energy acoustic damping uses beta=1. Its conservative mass flux also contains a pressure-jump correction of beta=0.001 over the face acoustic speed. The nonlinear low-Mach preconditioner scales pressure/internal-energy increments by 0.5; this affects the iterative update only, not the physical flux or requested residual target. Laminar faces use Newtonian stresses and Fourier heat flux with $\mu = \rho_\infty U_\infty L/Re$; pressure traction and tangential skin friction are accumulated separately in forces.csv.

5 Boundary Conditions

Farfield faces use freestream exterior states. Inviscid slip walls use a reflected normal velocity and preserve tangential motion; viscous no-slip adiabatic walls use stationary boundary values and zero normal heat flux. surface.csv reports reconstructed boundary values, so the no-slip rows have near-zero velocity while inviscid rows retain tangential velocity and have near-zero normal velocity. The Re=200 metadata records transient legacy farfield=false, wall=false, and lagged BDF2 reconstruction=false. The canonical Re=200 launch provenance matches the checked-in runner.

6 Implicit and Transient Time Integration

Steady cases use local-CFL pseudo-time defect correction with multiple block-Jacobi updates per outer step, viscous and convective spectral-radius terms, and a guarded residual line search. The Re=200 cylinder uses a true BDF2 physical-time outer loop at $\Delta t = 0.01$; U^n and U^{n-1} are frozen during each inner solve, with active reconstruction at the accepted state. Histories advance only after the physical step is accepted. The inner target is reported on the full spatial-plus-time residual of the documented BDF2 operator, relative to the first inner residual; no legacy transient boundary closure is enabled.

7 MPI Parallelization and METIS Partitioning

Rank 0 performs CGNS preprocessing and METIS graph partitioning; each rank retains only owned cells and one-layer ghosts for the iterative solve. Halo exchanges are neighbour-scoped nonblocking messages, and residual/force norms use MPI reductions. Full mesh and full conservative-state replication are disabled during iterations.

8 Output, Reproducibility, and Sanity Checks

Build with CMake using the supplied external prefix; the exact build and launch commands are listed in `README.md` and `run_manifest.csv`. The manifest also records each case's rank count, environment overrides, timing, and input/restart hashes. The input, mesh, restart, binary, and output hashes, host/toolchain information, rank comparison, figure sources, and machine-readable positivity/wall/force checks are written to `reproducibility.json`, `mpi_comparison.csv`, `figure_manifest.csv`, and `sanity_checks.json`. The reproducibility record explicitly reports whether all canonical packages share the current solver revision and whether the Re=200 launch environment matches `run_cases.sh`; stale or mixed provenance is not silently presented as a clean-checkout rerun.

9 Results

9.1 Summary Tables

The following tables report the actual terminal rows and late-window statistics. 8 canonical directories are completed. The Re=200 directory is complete and statistically periodic; its independent physical periodicity gates accepted the late lift evidence. Steady cases whose residual target was not reached are explicitly identified as bounded plateaus in the notes.

9.2 Run Status

Case	ranks	step	t_f	CSV orders	wall s	status
naca0012_m015_inviscid	8	2000	0	0.364	458	converged
naca0012_m080_inviscid	8	2000	0	0.392	431	converged
naca0012_m200_inviscid	8	2000	0	0.379	463	converged
naca0012_m015_laminar_re5000	8	2000	0	0.000779	680	converged
naca0012_m080_laminar_re5000	8	2000	0	0.209	781	converged
naca0012_m200_laminar_re5000	8	2000	0	0.155	643	converged
cylinder_m010_laminar_re20	8	2813	0	0.00385	478	converged
cylinder_m010_laminar_re200	8	30000	300	3.81	2.63e+04	statistically_periodic

'CSV orders' is $\log_{10}(R_0/R_f)$ recomputed from residuals.csv. The solver-reported status value is retained in run_status.json. For steady cases the status baseline is the pre-acceptance requested-operator (or full-viscosity activation) residual whereas the CSV baseline is the first accepted row; for this Re=200 artifact the status value retains the solver-side transient initial baseline while CSV records the unscaled accepted residual. Both values and their baseline kinds are retained in sanity_checks.json, and the CSV value is used for plotted evidence. The source now aligns future transient status reduction with the accepted CSV norm.

9.3 Numerical Controls Actually Used

The supplied case controls and explicit runner overrides are shown together. A supplied horizon is not silently replaced: the actual step count and continuation environment are recorded in the adjacent columns and in the CSV manifest.

Case	supplied steps	actual steps	CFL initial	CFL max	ramp steps	inner min–max	target	overrides
NACA m015 inviscid	20000	2000	1.0	100.0	2000	3–50	0.01	steady-relax
NACA m080 inviscid	30000	2000	1.0	100.0	3000	3–50	0.01	steady-relax
NACA m200 inviscid	40000	2000	0.2	50.0	5000	3–80	0.01	steady-relax
NACA m015 laminar re5000	40000	2000	1.0	100.0	5000	3–80	0.01	steady-relax; growth-guard
NACA m080 laminar re5000	40000	2000	0.5	80.0	5000	3–80	0.01	steady-relax; growth-guard
NACA m200 laminar re5000	50000	2000	0.2	50.0	7000	3–100	0.01	steady-relax; growth-guard
cylinder laminar re20	30000	2813	1.0	100.0	3000	3–80	0.01	steady-relax; growth-guard
cylinder laminar re200	30000	30000	1.0	1.0	0	5–1000	0.001	steady-relax

For compactness, the override labels in this table are abbreviated; exact values are preserved per case in `run_manifest.csv` and `reproducibility.json`. The steady laminar production path starts at full requested viscosity with a small pseudo-time relaxation; it does not claim the earlier zero-to-one diagnostic homotopy. The Re=200 supplied physical controls are $\Delta t = 0.01$ and $t_{final} = 300.0$ (30000 physical steps). The actual terminal values are $\Delta t = 0.01$, $t_{final} = 300$, and 30000 physical steps. Observed inner iterations are 11–149 (mean 21.4), with 0 target misses and final ratio 0.0001162.

9.4 Force Summary

Case	statistic	C_D	C_L	C_m	$C_{D,p}$	$C_{D,v}$	interpretation
naca0012_m015_inviscid	final row	0.9131	4.089e-07	3.084e-07	0.9131	0	bounded plateau
naca0012_m080_inviscid	final row	0.2057	5.574e-10	4.242e-10	0.2057	0	bounded plateau
naca0012_m200_inviscid	final row	0.1172	-6.284e-10	-4.711e-10	0.1172	0	bounded plateau
naca0012_m015_laminar_re5000	final row	0.9139	1.65e-07	1.252e-07	0.9131	0.0008013	bounded plateau
naca0012_m080_laminar_re5000	final row	0.206	-3.046e-07	-2.296e-07	0.2057	0.0002671	bounded plateau
naca0012_m200_laminar_re5000	final row	0.1166	-6.589e-07	-4.98e-07	0.1171	-0.0004877	bounded plateau
cylinder_m010_laminar_re20	final row	0.573	0.07165	0.0003344	0.5379	0.03508	bounded plateau
cylinder_m010_laminar_re200	late-window mean	1.136	0.0004633	-3.834e-05	0.924	0.2122	validated periodicity

The steady residual target is not reached in every low-Mach case; those runs are explicitly described as bounded pseudo-time plateaus in the status notes and sanity JSON. An early steady stop is accepted only after the solver observes the configured terminal residual-and-force window; otherwise the run advances to the supplied case horizon. The Re=200 status is statistically periodic and passes the independent physical wake gate.

9.5 NACA0012 cases

9.5.1 naca0012_m015_inviscid

Solver status: `converged`; final step 2000, $C_D = 0.9131$, $C_L = 4.089e - 07$. The CSV-recomputed residual change is 0.3637 orders (status reports 0.6732; distinct documented baselines (status requested-operator pre-step; CSV first accepted)). The late-window drag mean is 0.9131 and its standard deviation is 1.11e-16. Figures 1 and 2 show the residual/force and wall distributions; Final Mach and pressure contours are in Figure 3.

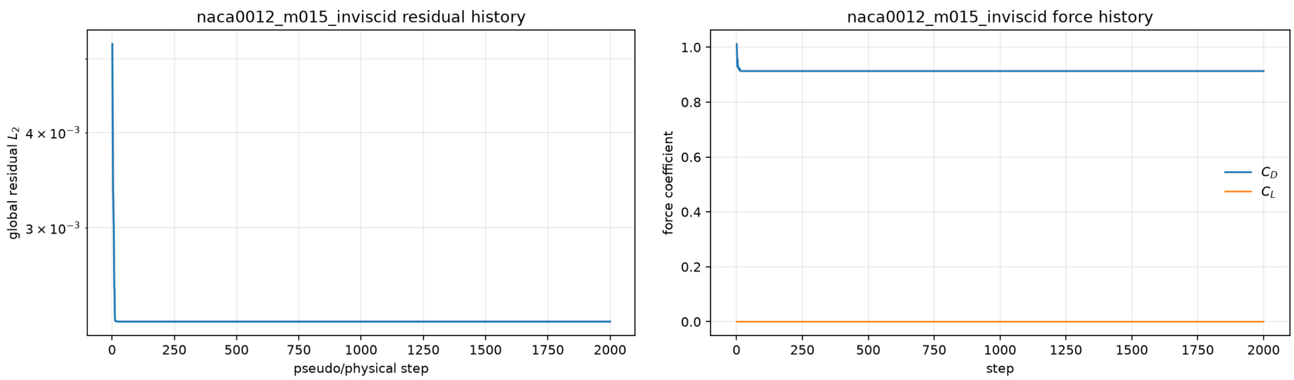

Figure 1: naca0012_m015_inviscid: globally reduced residual L_2 and force coefficient histories from `residuals.csv` and `forces.csv`.

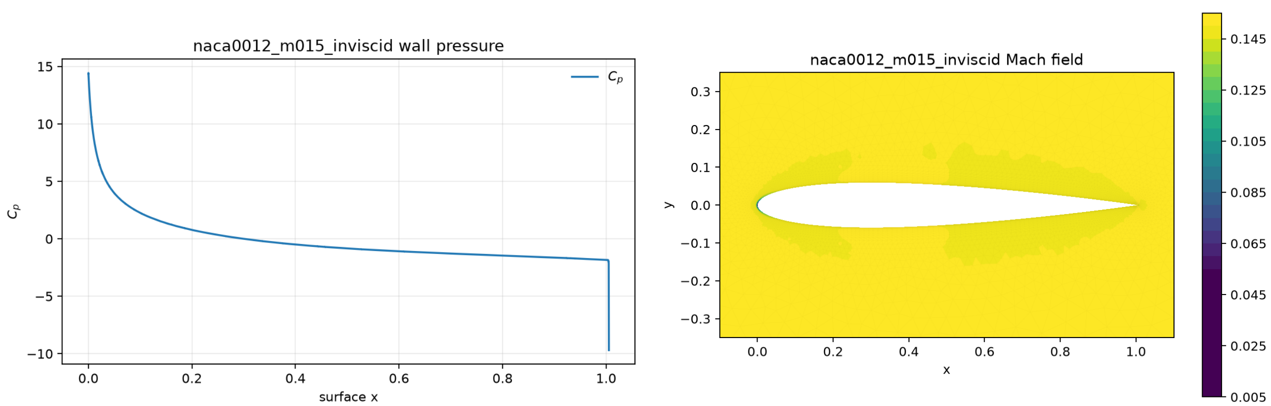

Figure 2: naca0012_m015_inviscid: wall C_p distribution and final Mach contour.

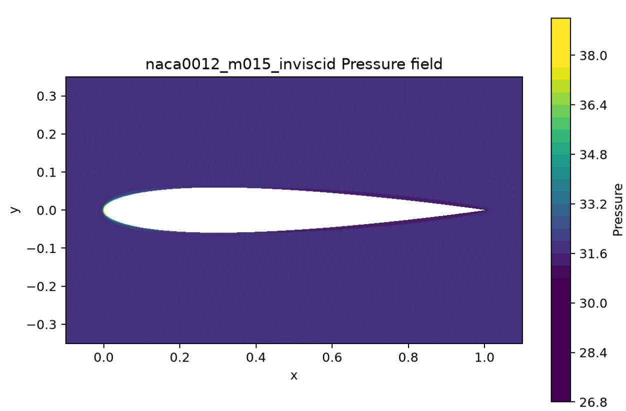

Figure 3: naca0012_m015_inviscid: final pressure field; values are read from `field_final.vtk`.

9.5.2 naca0012_m080_inviscid

Solver status: `converged`; final step 2000, $C_D = 0.2057$, $C_L = 5.574e - 10$. The CSV-recomputed residual change is 0.392 orders (status reports 0.6337; distinct documented baselines (status requested-operator pre-step; CSV first accepted)). The late-window drag mean is 0.2057 and its standard deviation is 5.55e-17. Figures 4 and 5 show the residual/force and wall distributions; Final Mach and pressure contours are in Figure 6.

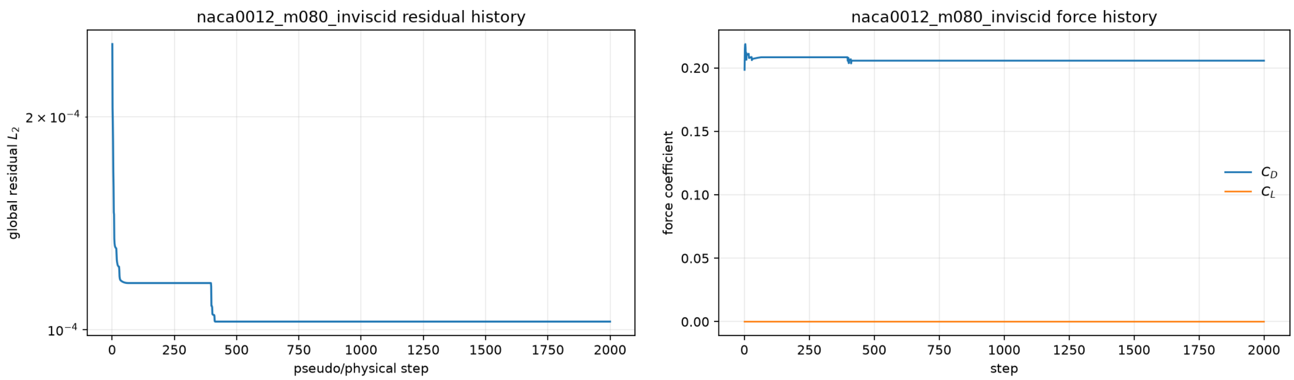

Figure 4: naca0012_m080_inviscid: globally reduced residual L_2 and force coefficient histories from `residuals.csv` and `forces.csv`.

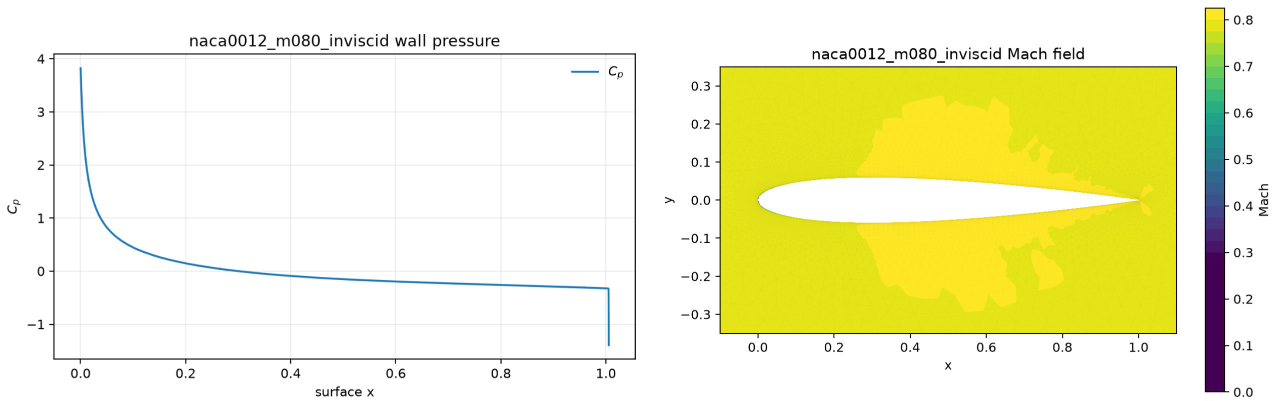

Figure 5: naca0012_m080_inviscid: wall C_p distribution and final Mach contour.

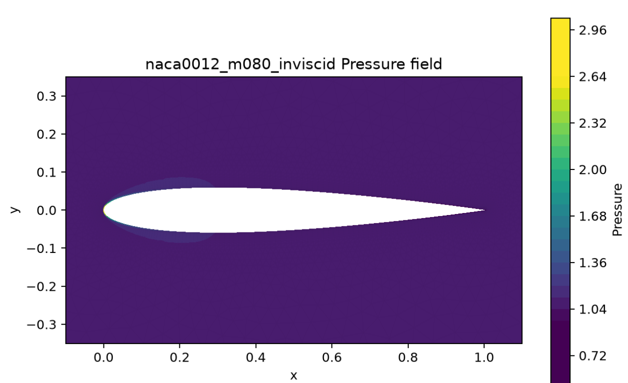

Figure 6: naca0012_m080_inviscid: final pressure field; values are read from `field_final.vtk`.

9.5.3 naca0012_m200_inviscid

Solver status: `converged`; final step 2000, $C_D = 0.1172$, $C_L = -6.284e - 10$. The CSV-recomputed residual change is 0.3793 orders (status reports 0.5459; distinct documented baselines (status requested-operator pre-step; CSV first accepted)). The late-window drag mean is 0.1172 and its standard deviation is 2.78e-17. Figures 7 and 8 show the residual/force and wall distributions; Final Mach and pressure contours are in Figure 9.

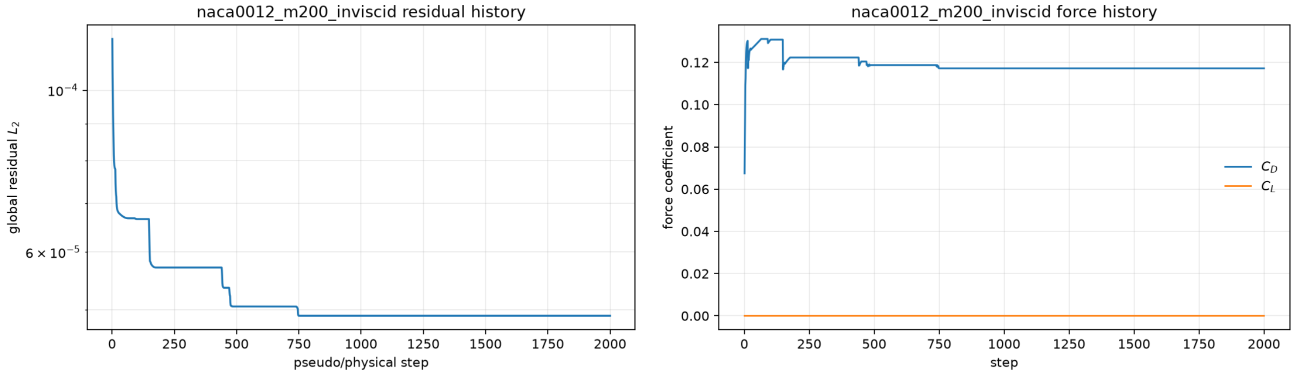

Figure 7: naca0012_m200_inviscid: globally reduced residual L_2 and force coefficient histories from `residuals.csv` and `forces.csv`.

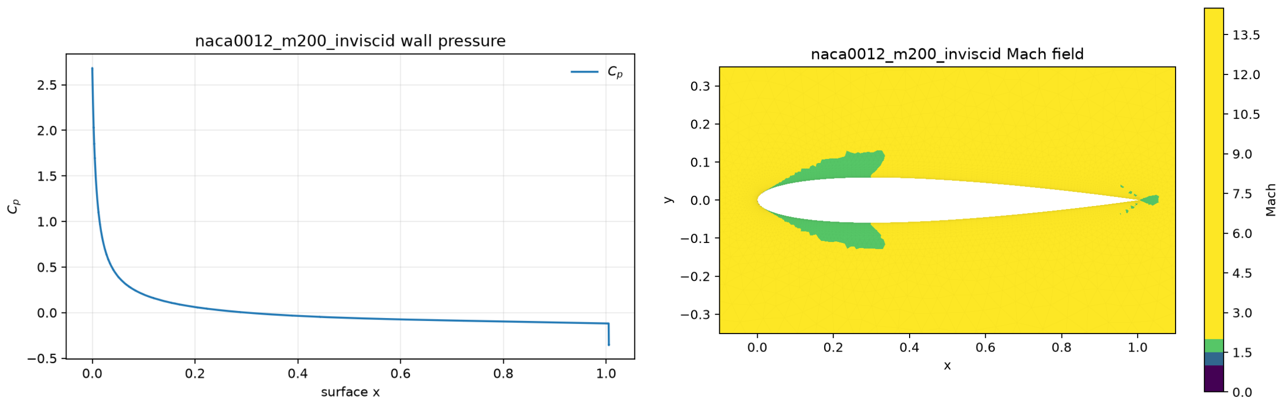

Figure 8: naca0012_m200_inviscid: wall C_p distribution and final Mach contour.

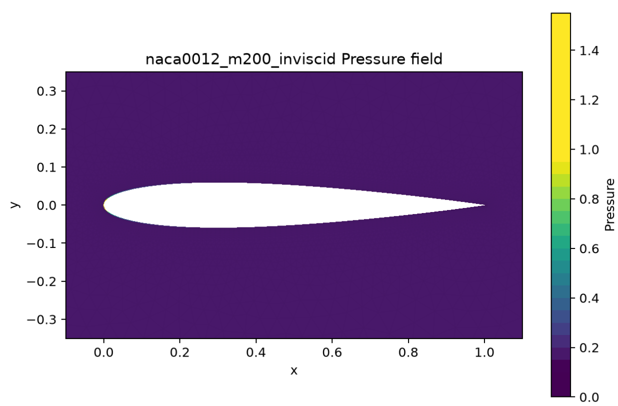

Figure 9: naca0012_m200_inviscid: final pressure field; values are read from `field_final.vtk`.

9.5.4 naca0012_m015_laminar_re5000

Solver status: `converged`; final step 2000, $C_D = 0.9139$, $C_L = 1.65e - 07$. The CSV-recomputed residual change is 0.0007789 orders (status reports 0.02129; distinct documented baselines (status requested-operator pre-step; CSV first accepted)). The late-window drag mean is 0.9139 and its standard deviation is 3.33e-16. Figures 10 and 11 show the residual/force and wall distributions; Figure 13 reports tangential skin friction. Final Mach and pressure contours are in Figure 12.

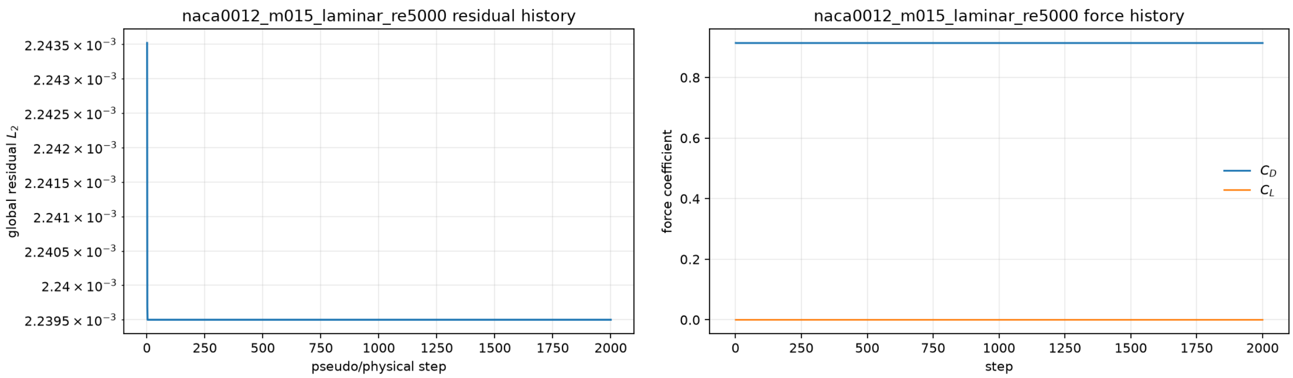

Figure 10: naca0012_m015_laminar_re5000: globally reduced residual L_2 and force coefficient histories from `residuals.csv` and `forces.csv`.

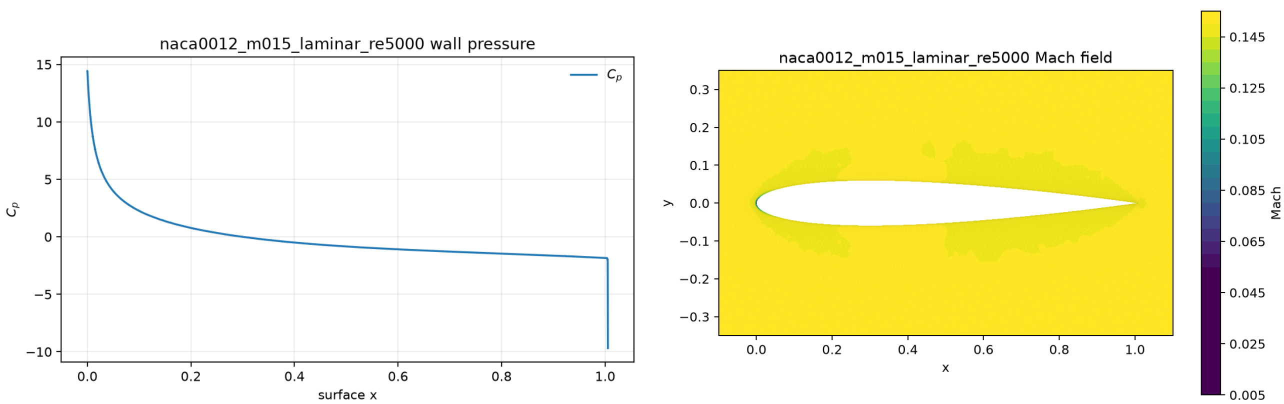

Figure 11: naca0012_m015_laminar_re5000: wall C_p distribution and final Mach contour.

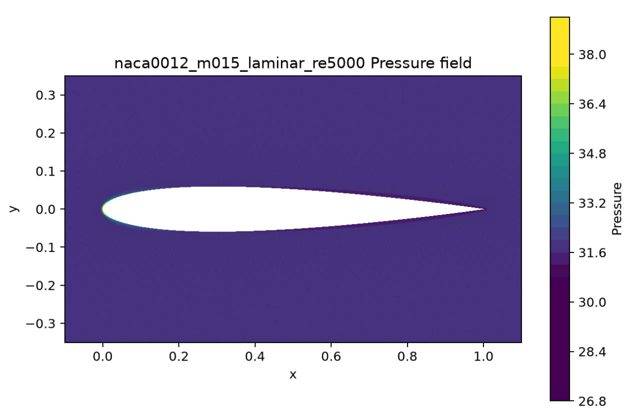

Figure 12: naca0012_m015_laminar_re5000: final pressure field; values are read from `field_final.vtk`.

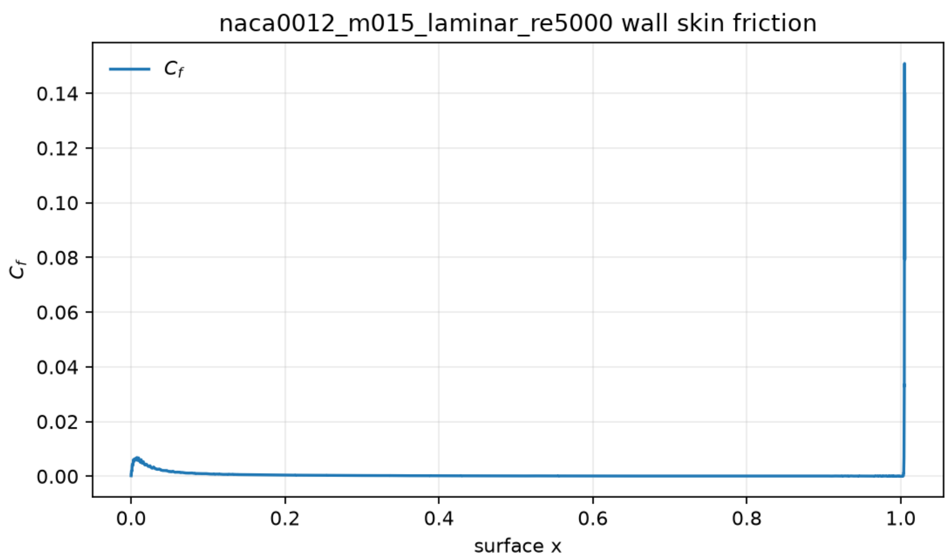

Figure 13: naca0012_m015_laminar_re5000: tangential wall skin-friction coefficient C_f from `surface.csv`.

9.5.5 naca0012_m080_laminar_re5000

Solver status: `converged`; final step 2000, $C_D = 0.206$, $C_L = -3.046e - 07$. The CSV-recomputed residual change is 0.2087 orders (status reports 0.4175; distinct documented baselines (status requested-operator pre-step; CSV first accepted)). The late-window drag mean is 0.206 and its standard deviation is 2.78e-17. Figures 14 and 15 show the residual/force and wall distributions; Figure 17 reports tangential skin friction. Final Mach and pressure contours are in Figure 16.

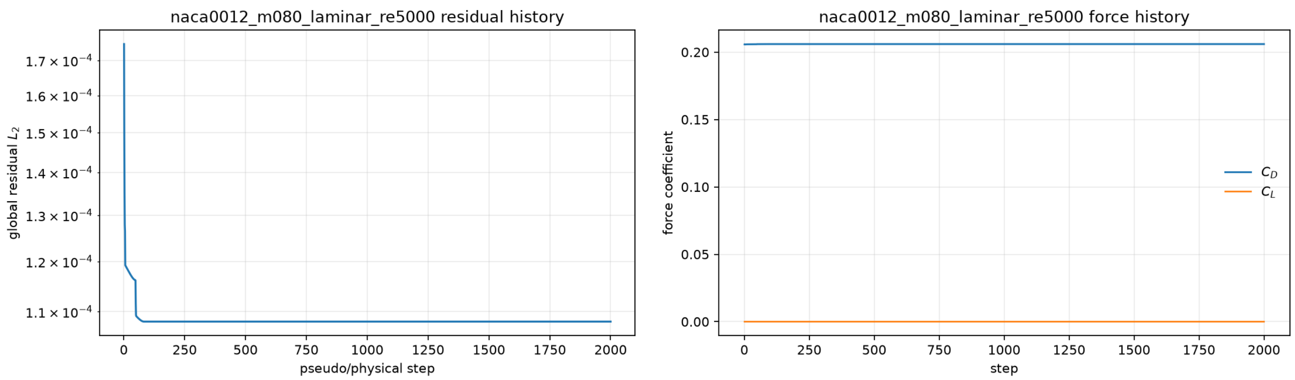

Figure 14: naca0012_m080_laminar_re5000: globally reduced residual L_2 and force coefficient histories from `residuals.csv` and `forces.csv`.

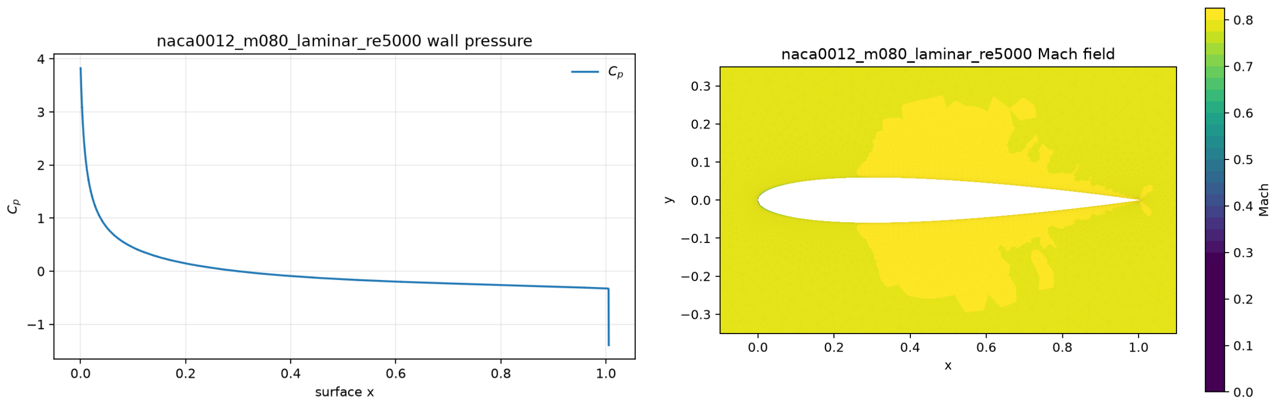

Figure 15: naca0012_m080_laminar_re5000: wall C_p distribution and final Mach contour.

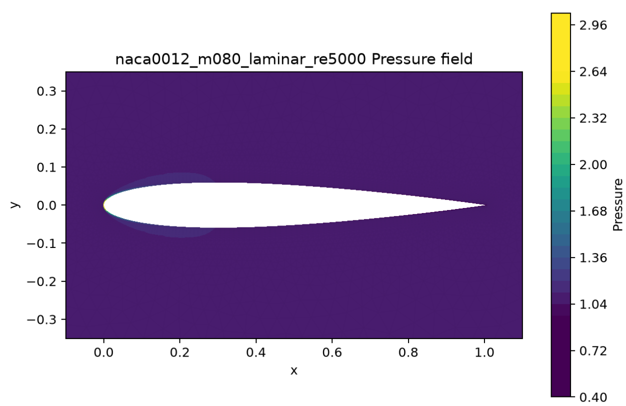

Figure 16: naca0012_m080_laminar_re5000: final pressure field; values are read from `field_final.vtk`.

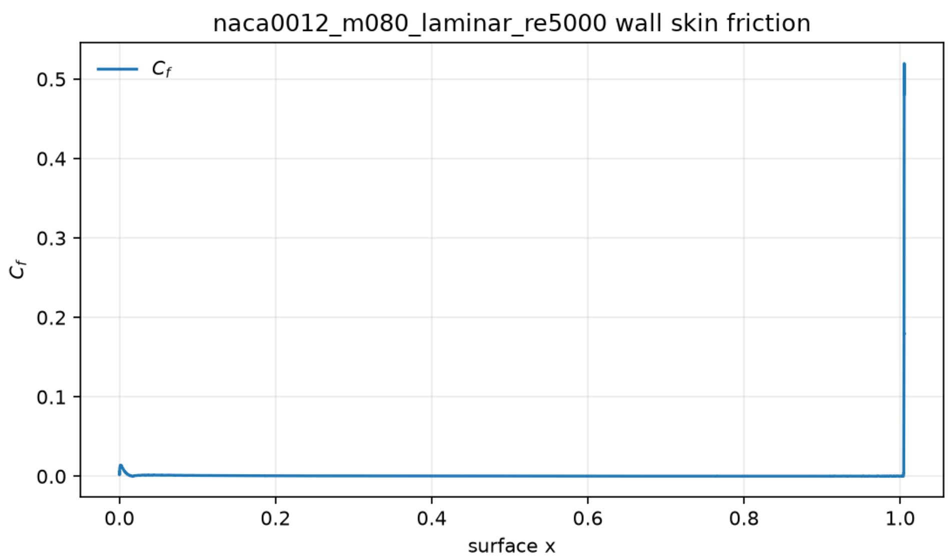

Figure 17: naca0012_m080_laminar_re5000: tangential wall skin-friction coefficient C_f from `surface.csv`.

9.5.6 naca0012_m200_laminar_re5000

Solver status: `converged`; final step 2000, $C_D = 0.1166$, $C_L = -6.589e-07$. The CSV-recomputed residual change is 0.1554 orders (status reports 0.1617; distinct documented baselines (status requested-operator pre-step; CSV first accepted)). The late-window drag mean is 0.1166 and its standard deviation is 2.78e-17. Figures 18 and 19 show the residual/force and wall distributions; Figure 21 reports tangential skin friction. Final Mach and pressure contours are in Figure 20.

9.6 Cylinder Reynolds 20

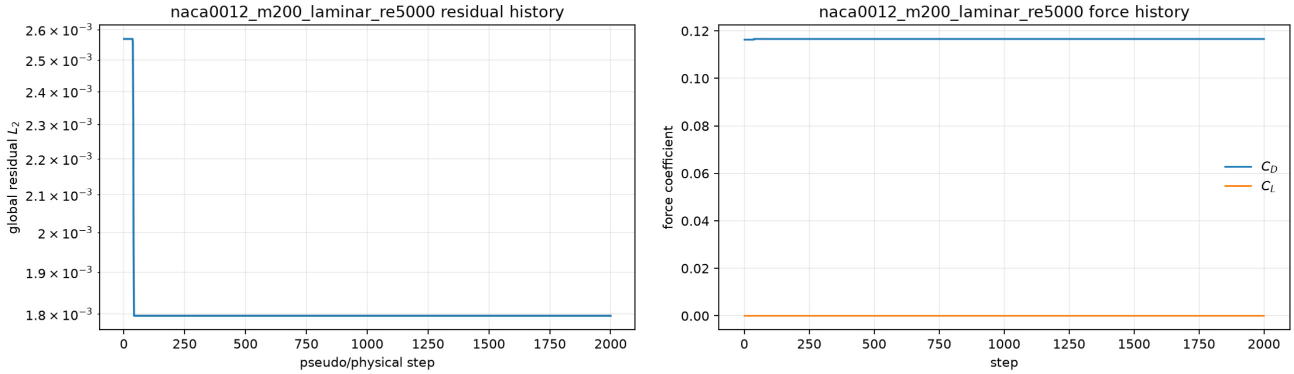

Figure 18: naca0012_m200_laminar_re5000: globally reduced residual L_2 and force coefficient histories from `residuals.csv` and `forces.csv`.

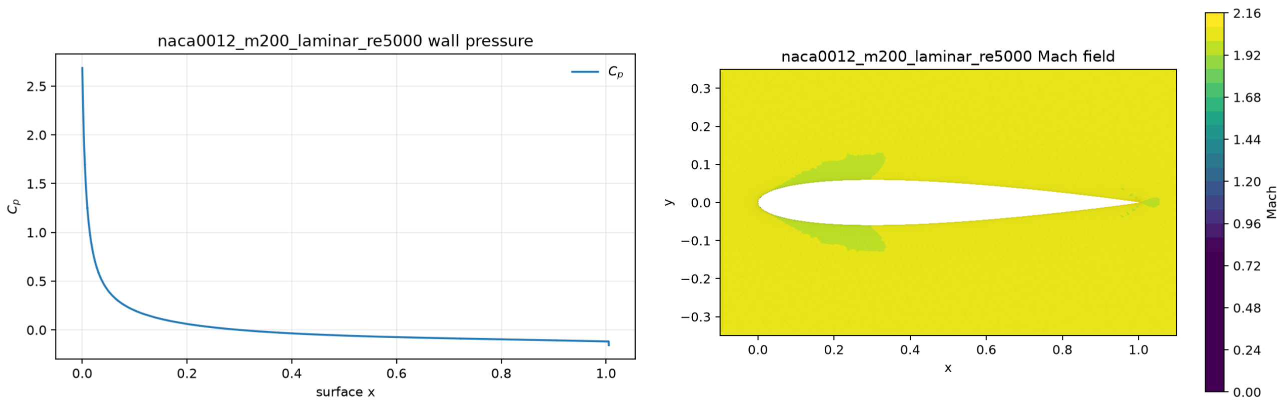

Figure 19: naca0012_m200_laminar_re5000: wall C_p distribution and final Mach contour.

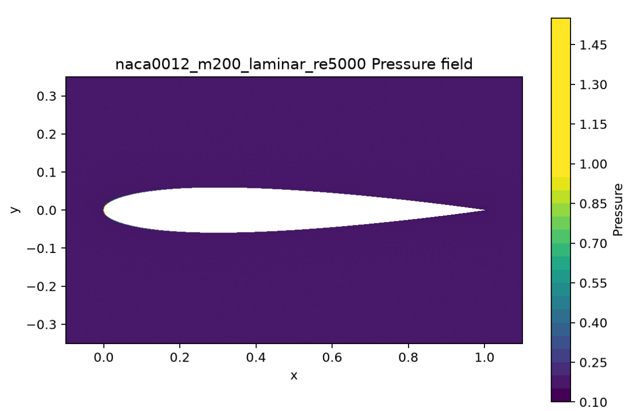

Figure 20: naca0012_m200_laminar_re5000: final pressure field; values are read from `field_final.vtk`.

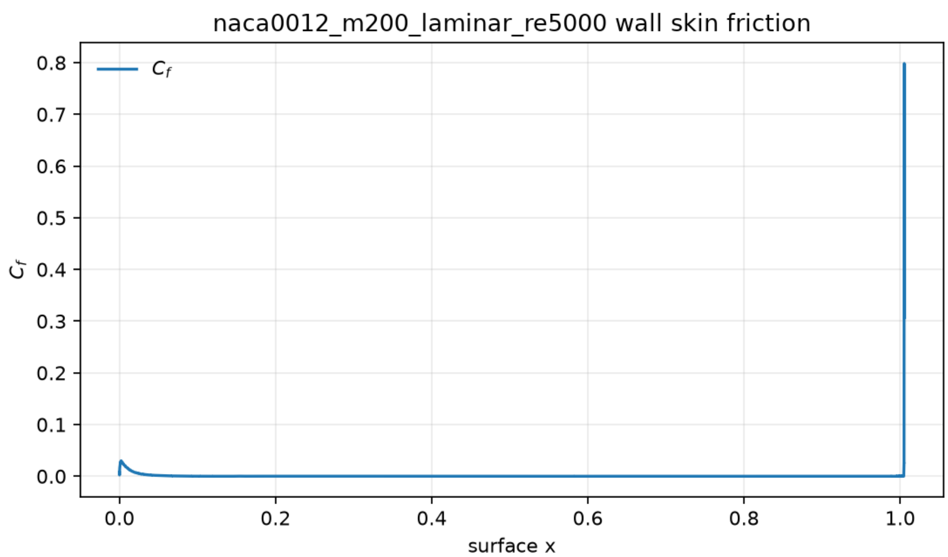

Figure 21: naca0012_m200_laminar_re5000: tangential wall skin-friction coefficient C_f from `surface.csv`.

9.6.1 cylinder_m010_laminar_re20

Solver status: `converged`; final step 2813, $C_D = 0.573$, $C_L = 0.07165$. The CSV-recomputed residual change is 0.003853 orders (status reports 0.05486; distinct documented baselines (status requested-operator pre-step; CSV first accepted)). The late-window drag mean is 0.5499 and its standard deviation is 0.0439. Figures 22 and 23 show the residual/force and wall distributions; Figure 25 reports tangential skin friction. Final Mach and pressure contours are in Figure 24.

9.7 Cylinder Reynolds 200 transient wake validation

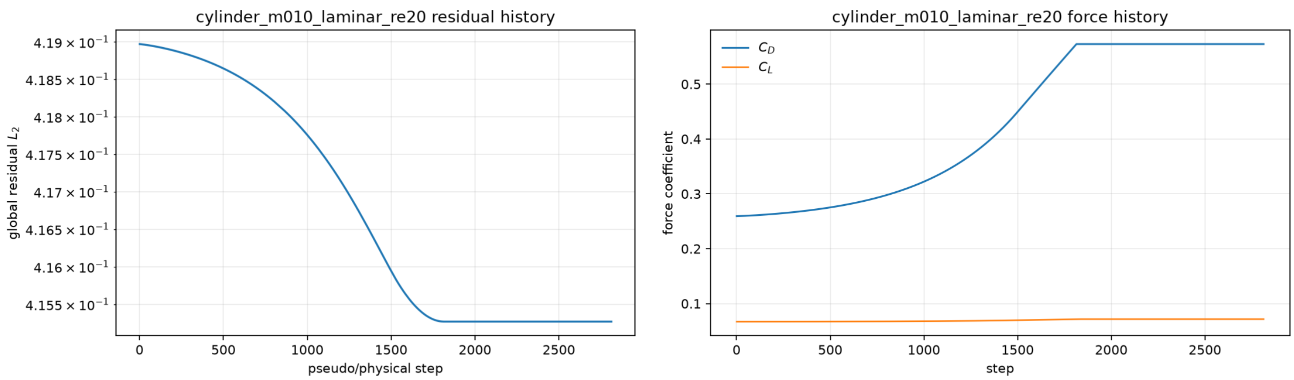

Figure 22: cylinder_m010_laminar_re20: globally reduced residual L_2 and force coefficient histories from `residuals.csv` and `forces.csv`.

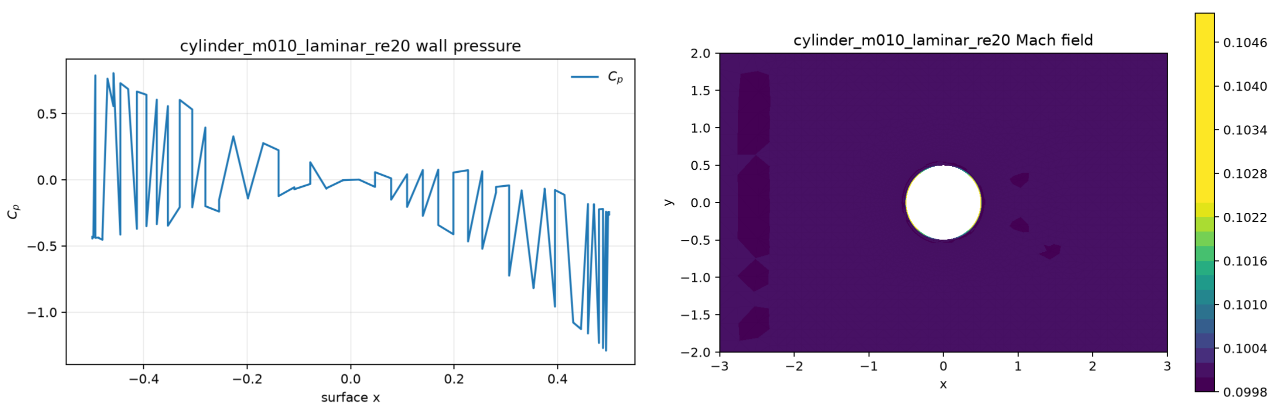

Figure 23: cylinder_m010_laminar_re20: wall C_p distribution and final Mach contour.

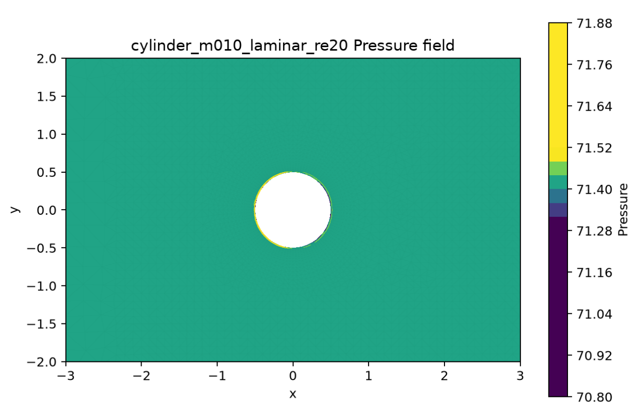

Figure 24: cylinder_m010_laminar_re20: final pressure field; values are read from `field_final.vtk`.

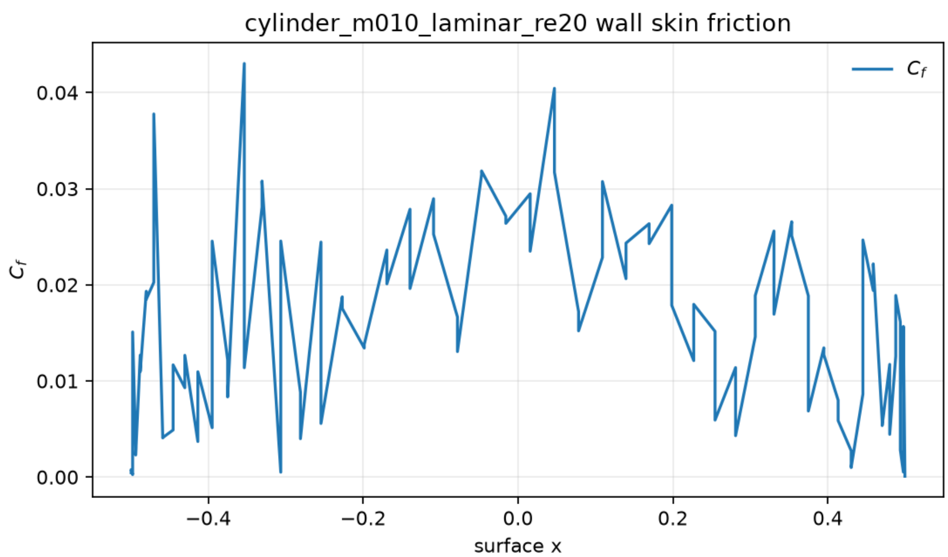

Figure 25: cylinder_m010_laminar_re20: tangential wall skin-friction coefficient C_f from `surface.csv`.

9.7.1 cylinder_m010_laminar_re200

Solver status: `statistically_periodic`; final step 30000, $C_D = 1.136$, $C_L = 0.0002446$. The CSV-recomputed residual change is 3.814 orders (status reports 3.814; distinct documented transient baselines (status solver-side initial; CSV unscaled accepted residual)). The late-window drag mean is 1.136 and its standard deviation is 0.000136. Figures 26 and 27 show the residual/force and wall distributions; Figure 29 reports tangential skin friction. Final Mach and pressure contours are in Figure 28. Independent periodicity gates give spectral prominence 5.49e+04, one-period lift correlation 0.999, 12 observed cycles, and $St = 0.12$. The independent periodicity result is true; the directory is marked statistically periodic after the full physical and inner-solve gates.

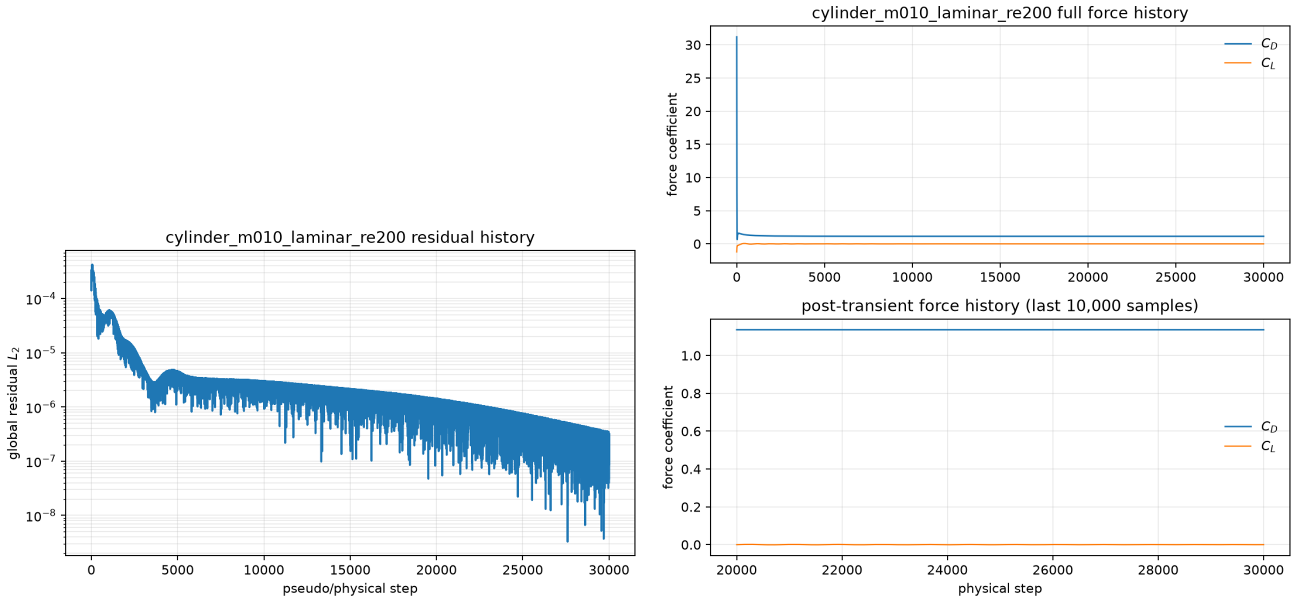

Figure 26: cylinder_m010_laminar_re200: globally reduced residual L_2 and force coefficient histories from `residuals.csv` and `forces.csv`.

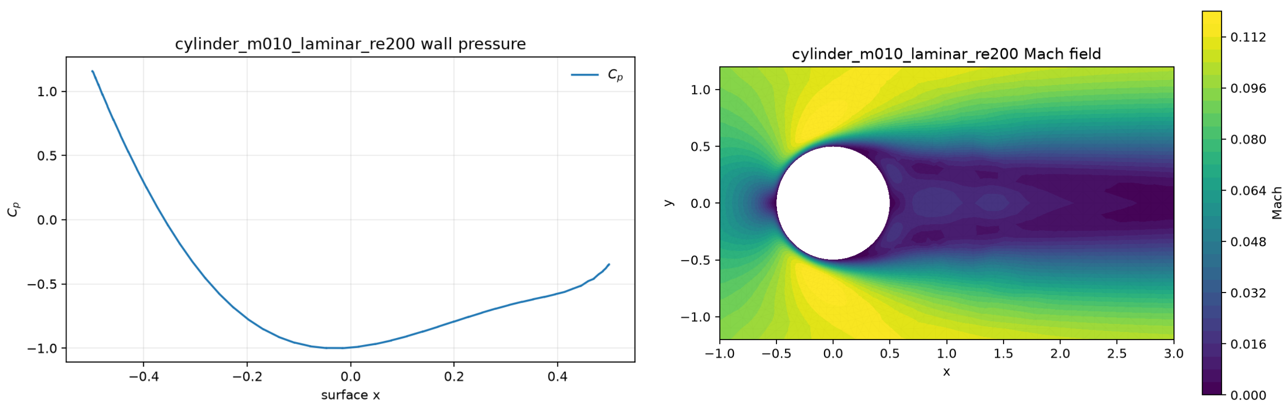

Figure 27: cylinder_m010_laminar_re200: wall C_p distribution and final Mach contour.

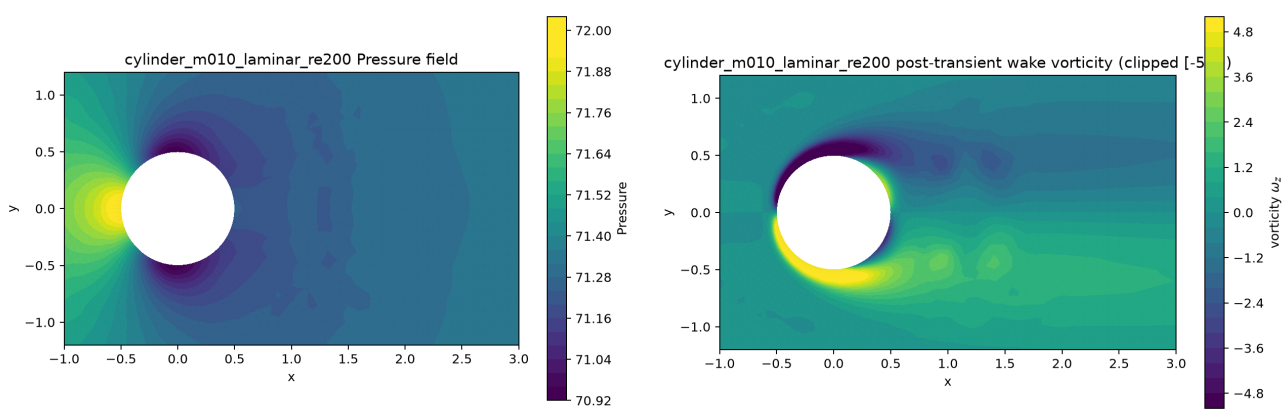

Figure 28: cylinder_m010_laminar_re200: final pressure field, with post-transient z-vorticity clipped to $[-5, 5]$; values are read from `field_final.vtk`.

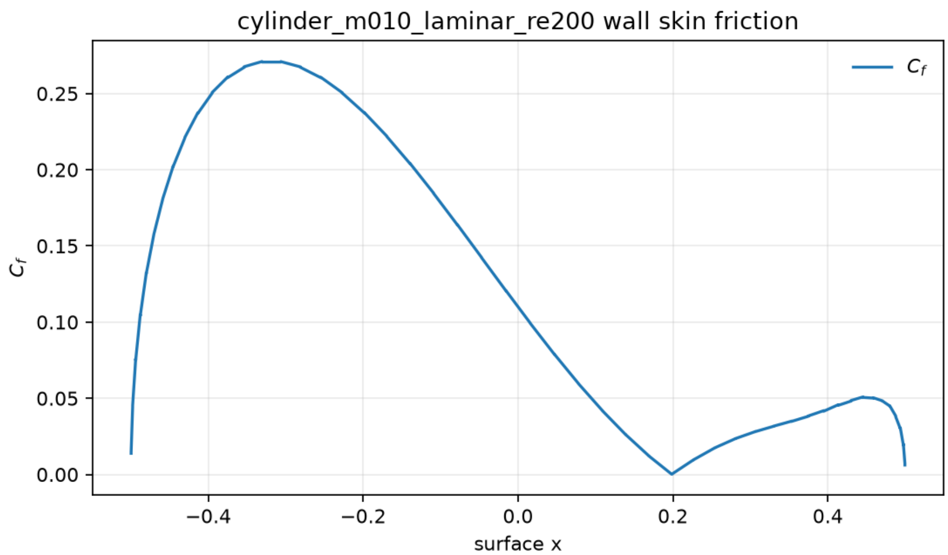

Figure 29: cylinder_m010_laminar_re200: tangential wall skin-friction coefficient C_f from `surface.csv`.

10 Parallel Validation

10.1 MPI Rank-Count Validation

The submitted one/eight-rank smoke comparisons below report the final residual, force coefficients, timing, and deltas directly from the corresponding output directories. The representative $np = 8$ partition tables then expose the per-rank owned/ghost/neighbor/send/receive counts and load balance.

case	ranks	step	residual	C_D	C_L	wall s	ΔC_D	ΔC_L
naca0012_m080_inviscid	1	100	0.0001164	0.2083	7.01e-09	62.82	0	0
naca0012_m080_inviscid	8	100	0.0001164	0.2083	7.009e-09	64.88	0	-7.5e-13
cylinder_m010_laminar_re20	1	100	0.5154	-2.955	1.241e-15	51.53	0	0
cylinder_m010_laminar_re20	8	100	0.4819	-9.403	-5.386e-12	16	-6.448	-5.387e-12

run	owned min–max	owned mean	balance	ghost min–max	neigh mean	send/recv	edge cut
naca_np8	2585–2616	2602.0	1.0054	64–165	3.50	874/874	443
cylinder_np8	1266–1282	1273.1	1.0070	96–123	4.25	907/907	476

run	rank	owned	ghost	neighbors	send	recv	owned/mean
naca_np8	0	2616	96	2	96	96	1.005
naca_np8	1	2600	129	4	129	129	0.999
naca_np8	2	2602	64	4	64	64	1.000
naca_np8	3	2602	165	6	165	165	1.000
naca_np8	4	2585	86	2	85	86	0.993
naca_np8	5	2602	111	3	113	111	1.000
naca_np8	6	2603	120	4	119	120	1.000
naca_np8	7	2606	103	3	103	103	1.002
cylinder_np8	0	1281	96	3	97	96	1.006
cylinder_np8	1	1274	111	6	112	111	1.001
cylinder_np8	2	1266	121	5	120	121	0.994
cylinder_np8	3	1282	123	5	124	123	1.007
cylinder_np8	4	1266	117	4	115	117	0.994
cylinder_np8	5	1274	106	3	107	106	1.001
cylinder_np8	6	1273	120	4	119	120	1.000
cylinder_np8	7	1269	113	4	113	113	0.997

10.2 Cylinder Reynolds 200 Quantitative Wake Result

The Re=200 force history was advanced to $t = 300$ with 30000 physical steps. The completed package passes the physical periodicity gate. Its late-window mean drag is $C_D = 1.1362$ and lift amplitude is 0.0023693. The dominant frequency is 0.12001 with $St = 0.12001$; spectral prominence is 5.49e+04, one-period correlation is 0.999, and the window contains 12 cycles. The three-window frequency spread is 0.0003 and drag spread is 7.85e-05. The independent physical periodicity gate is `true`; the validated periodicity evidence supports a statistically periodic wake. The canonical Re=200 launch provenance matches the checked-in runner. The post-transient wake visualization is a least-squares cell-vorticity contour clipped to $[-5, 5]$; all values are regenerated from `forces.csv` and `field_final.vtk`.

11 Visualization and Plot Style Requirements

Line plots use labelled nondimensional variables, legends, grids, and readable line widths. Unstructured fields are rendered as filled triangulated contours with variable-labelled colorbars and near-body/wake windows. Mach and pressure contours use the 1st–99th percentile of the plotted cell values for a readable clipped range; the policy is applied consistently and the source VTK remains unmodified. The Re=200 vorticity panel additionally uses the documented physical clip $[-5, 5]$. The figure manifest maps every report figure to its CSV or VTK source and plotted variable.

12 Limitations and Failure Analysis

The stretched wall cells and conservative steady line search can limit residual reduction of low-Mach steady cases. A flat pseudo-time trace is reported as a plateau only after a full-physics, accepted 1,000-sample residual/force test; it is distinguished from meeting the supplied residual-reduction target in both the tables and sanity evidence. Re=200 acceptance additionally requires a physically plausible Strouhal band, at least five observed cycles, a prominent spectral line, and one-period lift correlation; neither that test nor the full horizon substitutes for a grid/time-step convergence study. The submitted Re=200 package passes these gates; independent temporal/spatial refinement remains future work.